

CO2_Assesment 1

1. Employee & Department

```
mysql> CREATE TABLE Department (  
-> DeptID INT PRIMARY KEY,  
-> DeptName VARCHAR(50) NOT NULL  
-> );  
ERROR 1046 (3D000): No database selected  
mysql> use simats;  
Database changed  
mysql> CREATE TABLE Department (  
-> DeptID INT PRIMARY KEY,  
-> DeptName VARCHAR(50) NOT NULL  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> CREATE TABLE Employee (  
-> EmpID INT PRIMARY KEY,  
-> Name VARCHAR(50) NOT NULL,  
-> Salary DECIMAL(10,2),  
-> DeptID INT,  
-> FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
-> );  
Query OK, 0 rows affected (0.10 sec)  
  
mysql> INSERT INTO Department VALUES (10, 'IT'), (20, 'HR'), (30, 'Finance');  
Query OK, 3 rows affected (0.03 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO Employee VALUES  
-> (1, 'Arun', 60000, 10),  
-> (2, 'Bala', 45000, 10),  
-> (3, 'Chitra', 50000, 20),  
-> (4, 'Divya', 70000, 30);  
Query OK, 4 rows affected (0.03 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

2. Student & Department

```
mysql> create table dept(  
-> DeptID INT PRIMARY KEY,  
-> DeptName VARCHAR(50) UNIQUE NOT NULL  
-> );  
Query OK, 0 rows affected (0.08 sec)  
  
mysql> CREATE TABLE Student (  
-> SID INT PRIMARY KEY,  
-> Name VARCHAR(50) NOT NULL,  
-> DOB DATE NOT NULL,  
-> DeptID INT,  
-> FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
-> );  
Query OK, 0 rows affected (0.08 sec)  
  
mysql> INSERT INTO Department VALUES (1, 'CSE'), (2, 'ECE'), (3, 'MECH');  
Query OK, 3 rows affected (0.03 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO Student VALUES  
-> (101, 'Karthik', '2002-05-10', 1),  
-> (102, 'Priya', '2001-08-15', 2),  
-> (103, 'Ravi', '2003-01-20', 1);  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

3. Sales Dataset

```

mysql> CREATE TABLE Sales (
    ->     SaleID INT PRIMARY KEY,
    ->     ProductID INT,
    ->     Region VARCHAR(50),
    ->     Amount DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> -- Sample Data
mysql> INSERT INTO Sales VALUES
    -> (1, 100, 'North', 5000),
    -> (2, 100, 'South', 7000),
    -> (3, 200, 'North', 12000),
    -> (4, 300, 'East', 8000),
    -> (5, 200, 'West', 15000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

```

4. Correlated Subquery – Employee Salary vs Dept Avg

```

mysql> SELECT E.EmpID, E.Name, E.Salary, E.DeptID
    -> FROM Employee E
    -> WHERE E.Salary > (
    ->     SELECT AVG(Salary)
    ->     FROM Employee
    ->     WHERE DeptID = E.DeptID
    -> );

```

EmpID	Name	Salary	DeptID
1	Arun	60000.00	10

```

1 row in set (0.02 sec)

```

5. Project Table (for JOINS)

```

mysql> CREATE TABLE Project (
->     ProjectID INT PRIMARY KEY,
->     ProjectName VARCHAR(50),
->     EmpID INT,
->     FOREIGN KEY (EmpID) REFERENCES Employee(EmpID)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- Sample Data
mysql> INSERT INTO Project VALUES
-> (501, 'AI System', 1),
-> (502, 'Payroll', 3),
-> (503, 'Audit', 4);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select *from Project;
+-----+-----+-----+
| ProjectID | ProjectName | EmpID |
+-----+-----+-----+
|      501 | AI System   |      1 |
|      502 | Payroll     |      3 |
|      503 | Audit       |      4 |
+-----+-----+-----+
3 rows in set (0.00 sec)

```

6. DeptSalaryView – View Creation

```

mysql> CREATE VIEW DeptSalaryView AS
-> SELECT D.DeptName, COUNT(E.EmpID) AS TotalEmployees, AVG(E.Salary) AS
AvgSalary
-> FROM Department D
-> JOIN Employee E ON D.DeptID = E.DeptID
-> GROUP BY D.DeptName;
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> -- Query on view
mysql> SELECT * FROM DeptSalaryView WHERE AvgSalary > 50000;
+-----+-----+-----+
| DeptName | TotalEmployees | AvgSalary |
+-----+-----+-----+
| IT       | 2              | 52500.000000 |
| Finance  | 1              | 70000.000000 |
+-----+-----+-----+
2 rows in set (0.00 sec)

```

7. Course & Enrollment

```

mysql> CREATE TABLE Course (
  ->     CourseID INT PRIMARY KEY,
  ->     CourseName VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE Enrollment (
  ->     SID INT,
  ->     CourseID INT,
  ->     PRIMARY KEY (SID, CourseID),
  ->     FOREIGN KEY (SID) REFERENCES Student(SID),
  ->     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- Sample Data
mysql> INSERT INTO Course VALUES
  -> (201, 'DBMS'),
  -> (202, 'OS'),
  -> (203, 'Networks');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Enrollment VALUES
  -> (101, 201), (101, 202), (101, 203),
  -> (102, 201), (102, 202),
  -> (103, 201), (103, 202), (103, 203);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

```

7. Course & Enrollment

```

mysql> -- Create Learner table
mysql> CREATE TABLE Learner (
    ->     SID INT PRIMARY KEY,
    ->     Name VARCHAR(50) NOT NULL,
    ->     DOB DATE NOT NULL,
    ->     DeptID INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> -- Insert sample data
mysql> INSERT INTO Learner (SID, Name, DOB, DeptID)
    -> VALUES
    -> (101, 'Arun', '2002-05-10', 1),
    -> (102, 'Priya', '2001-08-15', 2),
    -> (103, 'Ravi', '2003-01-20', 1);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- Update example
mysql> UPDATE Learner
    -> SET DeptID = 2
    -> WHERE SID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> -- Delete example
mysql> DELETE FROM Learner
    -> WHERE SID = 101;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> -- Check remaining data
mysql> SELECT * FROM Learner;
+-----+-----+-----+-----+
| SID | Name  | DOB       | DeptID |
+-----+-----+-----+-----+
| 102 | Priya | 2001-08-15 | 2      |
| 103 | Ravi  | 2003-01-20 | 1      |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

8. DML with Constraints

```

mysql> CREATE TABLE Scholar (
    ->     SID INT PRIMARY KEY,
    ->     Name VARCHAR(50) NOT NULL,
    ->     DOB DATE NOT NULL,
    ->     DeptID INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- Insert sample data
mysql> INSERT INTO Scholar (SID, Name, DOB, DeptID)
    -> VALUES
    -> (101, 'Arun', '2002-05-10', 1),
    -> (102, 'Priya', '2001-08-15', 2),
    -> (103, 'Ravi', '2003-01-20', 1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- Update example
mysql> UPDATE Scholar
    -> SET DeptID = 2
    -> WHERE SID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> -- Delete example
mysql> DELETE FROM Scholar
    -> WHERE SID = 101;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> -- Check remaining data
mysql> SELECT * FROM Scholar;
+-----+-----+-----+-----+
| SID | Name  | DOB       | DeptID |
+-----+-----+-----+-----+
| 102 | Priya | 2001-08-15 | 2      |
| 103 | Ravi  | 2003-01-20 | 1      |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

9. Tuple Relational Calculus

```

mysql> CREATE TABLE Staff (
->     StaffID INT PRIMARY KEY,
->     Name VARCHAR(50) NOT NULL,
->     DeptName VARCHAR(50) NOT NULL,
->     Salary DECIMAL(10,2) NOT NULL
-> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> -- Insert sample data
mysql> INSERT INTO Staff (StaffID, Name, DeptName, Salary)
-> VALUES
-> (1, 'Arun', 'IT', 60000),
-> (2, 'Bala', 'IT', 45000),
-> (3, 'Chitra', 'HR', 52000),
-> (4, 'Divya', 'Finance', 70000),
-> (5, 'Ravi', 'IT', 55000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> -- Query: Find all staff in IT earning > 50000
mysql> SELECT *
-> FROM Staff
-> WHERE DeptName = 'IT' AND Salary > 50000;
+-----+-----+-----+-----+
| StaffID | Name | DeptName | Salary |
+-----+-----+-----+-----+
|      1 | Arun | IT      | 60000.00 |
|      5 | Ravi | IT      | 55000.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

10. Airline Reservation System – Schema & Queries

```
mysql> CREATE TABLE Passenger (
->     PassengerID INT PRIMARY KEY,
->     Name VARCHAR(50),
->     Contact VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE Flight (
->     FlightID INT PRIMARY KEY,
->     Source VARCHAR(50),
->     Destination VARCHAR(50),
->     DateTime DATETIME
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE Reservation (
->     ResID INT PRIMARY KEY,
->     PassengerID INT,
->     FlightID INT,
->     SeatNo VARCHAR(10),
->     FOREIGN KEY (PassengerID) REFERENCES Passenger(PassengerID),
->     FOREIGN KEY (FlightID) REFERENCES Flight(FlightID)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE Payment (
->     PaymentID INT PRIMARY KEY,
->     ResID INT,
->     Amount DECIMAL(10,2),
->     Status VARCHAR(20),
->     FOREIGN KEY (ResID) REFERENCES Reservation(ResID)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- Sample Queries
mysql> -- 1. Join Passenger & Reservation
mysql> SELECT P.Name, R.SeatNo
-> FROM Passenger P JOIN Reservation R ON P.PassengerID = R.PassengerID;
Empty set (0.00 sec)
```



```
mysql> -- 2. Flights with > 100 reservations
mysql> SELECT FlightID
      -> FROM Reservation
      -> GROUP BY FlightID
      -> HAVING COUNT(*) > 100;
Empty set (0.00 sec)

mysql>
mysql> -- 3. View
mysql> CREATE VIEW FlightBookingView AS
      -> SELECT F.FlightID, COUNT(R.ResID) AS TotalBookings
      -> FROM Flight F JOIN Reservation R ON F.FlightID = R.FlightID
      -> GROUP BY F.FlightID;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> -- 4. Query on view
mysql> SELECT * FROM FlightBookingView WHERE TotalBookings > 50;
Empty set (0.00 sec)

mysql>
mysql> -- 5. Correlated subquery
mysql> SELECT P.Name
      -> FROM Passenger P
      -> WHERE EXISTS (
      ->     SELECT 1 FROM Reservation R
      ->     WHERE R.PassengerID = P.PassengerID
      ->     GROUP BY R.PassengerID
      ->     HAVING COUNT(*) > 1
      -> );
Empty set (0.00 sec)
```